WHITEPAPER V0.2 · TARGET FINAL DESIGN

GumBall6900

A signal-directed onchain fund

Holders mine GBX from a fixed public schedule, stake it into non-transferable sGBX, and continuously direct where the fund's next dollar goes. Acquisitions clear against the open market. Redemption pays out real tokens, in kind. The deployed core is fixed: signals govern capital, code governs the rules.

1,000,000,000	**98%**	**Zero**	**Five**
GBX LIFETIME CEILING	PUBLIC MINING CAPACITY	TEAM OR INVESTOR ALLOCATION	MANAGEMENT ACTIONS

August 2026

Pre-audit · not deployed · not authorized for user funds

READ THIS FIRST

This describes a design, not a deployed system.

GumBall6900 is an experimental, pre-audit protocol design. It is not deployed, not audited, and not authorized for user funds. Nothing here is investment, legal, or tax advice.

This paper specifies the intended governance-minimized final system. The contracts currently in the repository still expose a broader administrative surface than the design described here, and that gap must be closed and independently reviewed before any deployment. Where this paper and deployed bytecode ever disagree, the bytecode controls behaviour.

VERSION

v0.2, August 2026. Supersedes v0.1. Figures are generated from the same economic model the repository tests.

COLOUR IS NOT DECORATION

Every figure uses the same grammar. A line's colour tells you what is moving — and pink traces the whole holder-directed chain, from the signal through the acquisition it causes to the reward it pays.

- **USDG capital** — Contributed and routed in
- **Signal, and what it buys** — Weight, acquisition and reward
- **GBX supply and burns** — Minted once, burned forever

THREE WAYS THROUGH

The mechanism

Read the figure on the next page, then sections 3 to 12.

The economics

Sections 5, 6 and 13, then the worked example in section 14.

What can go wrong

Sections 16 to 18 state what must hold, what can fail, and what is unfinished.

FIXED BY CONSTRUCTION

Set before deployment. Only the signal-reward share can be changed afterwards, and only within its cap.

1,000,000,000 GBX
Cumulative lifetime mint ceiling

20,000,000 GBX
Genesis liquidity, committed

0 GBX
Team, investor and presale allocation

1,460 days
Emission half-life

10%
Signal-reward share at launch, capped at 50%

None
Withdrawal lock, cooldown or epoch

HOW THIS PAPER IS ARRANGED

Contents

The design first, then the economics, then an honest account of the risks and the unfinished work.

WHERE THE NUMBERS COME FROM

Every chart is computed at build time from the emission, auction and redemption rules in `packages/simulations`, so a figure cannot drift away from the model the repository tests.

NAMING

Contract names are capitalised — Fundraiser, Resonance, Strategy, Bribe, Fund. Tokens are set in their tickers: GBX, sGBX, USDG.

ORIENTATION

The protocol in one figure

Three kinds of value move through GumBall6900. This is all of them, and every place they can end up.

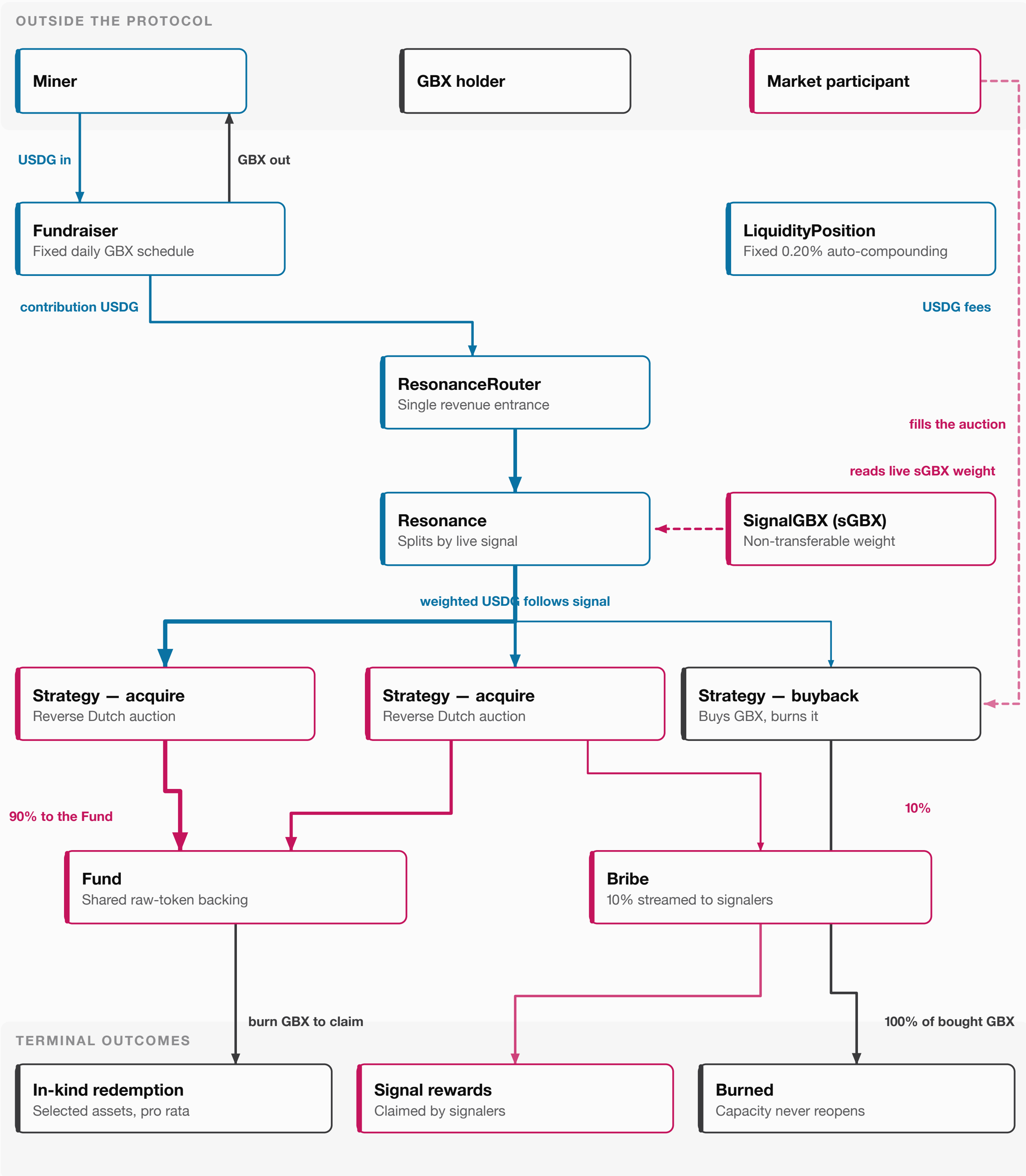


FIG. 01 Contribution revenue has exactly one entrance and follows live signal weight to a Strategy. Acquisitions divide between shared backing and the signalers who directed them, 90/10 at launch. Redemption and burning are the only exits, and burned capacity never reopens.

FROM INDEX PRODUCT TO INDEX FORMATION

1 The problem

Onchain funds can move a basket onchain while leaving the decision of what belongs in it somewhere else entirely.

A traditional fund makes a basket easier to own. It does not make the basket easier to influence. Membership, weighting and rebalancing are decided upstream by a methodology or a committee, and the investor receives a finished result. Many onchain index products preserve exactly this shape: the token is onchain, but formation of the basket remains a separate, off-chain process.

GumBall6900 starts from a narrower question. Instead of asking who should decide the portfolio, it asks who should decide the *next purchase*.

> **What if holders continuously directed where the fund's next dollar went?**

That change moves the unit of coordination. Holders never approve a complete target portfolio, and there is no proposal to pass or reject. They hold live weight across the eligible acquisition paths, and every new unit of protocol revenue follows whatever distribution exists at the moment it is routed.

The consequence runs through the whole design: the Fund is a record of past acquisitions, while signals are a statement of present preference. Changing a signal changes what happens next. It does not sell what the protocol already holds.

THE DISTINCTION TO KEEP

GBX is a claim on what the Fund already owns. **sGBX** directs what it may acquire next. They are deliberately different instruments.

NO FORCED SELLING

Because signals steer inflow rather than holdings, a swing in sentiment cannot force the Fund to liquidate. Section 13 shows what that means for composition over time.

CONVENTIONAL INDEX PRODUCT

The basket arrives finished

1. A methodology sets membership
2. A committee sets weights
3. Investors receive the result

vs

GUMBALL6900

The basket is formed continuously

1. Strategies define eligible paths
2. Holders signal where flow goes
3. Each acquisition adds to the basket

FIG. 02 The same three decisions, relocated. Nothing about the second column requires a vote, a season, or a quorum.

WHAT THE SYSTEM IS TRYING TO PRESERVE

2 Design goals

Six commitments shape the design. Each one costs something, and the cost is stated alongside it.

GOAL	WHAT IT MEANS IN PRACTICE
Fair public distribution	GBX is mined through the Fundraiser. There is no team, founder, investor, presale or advisor allocation; the separate 20M genesis tranche is committed to canonical liquidity.
Continuous direction	Absolute per-Strategy sGBX amounts may be increased or decreased at any time. No voting season, allocation epoch or signal cooldown; unallocated sGBX is immediately withdrawable.
Real assets	The Fund holds raw tokens and redemption transfers them in kind, so the core never needs a protocol-wide net asset value oracle.
Market-based acquisition	Reverse Dutch auctions let external participants decide when a price is acceptable, rather than the protocol asserting one.
Governance minimization	Signals govern capital. Resonance holds four explicitly authorized actions, each bounded, and the deployed core cannot be upgraded.
Inspectability	Each contract has a narrow responsibility, so money flow and failure boundaries stay legible to a reader who is not the author.

THE COST OF EACH

Fair distribution means no funded runway. Continuous direction means no deliberation. Immutability means no fix. These are trades, not free wins.

Non-goals

THE PROTOCOL DOES NOT PROMISE

- A stable or rising GBX price.
- A fixed portfolio composition or automatic rebalancing.
- That any auction will clear, or clear well.
- That signaling earns a reward.
- Protection from the risks of assets the Fund receives.

THE CONTRACTS DO FIX

- That every contributed USDG enters the signal-directed path.
- That miners compete for a schedule, not a team-set price.
- That the Fund keeps the majority of every acquisition, whatever the reward share is set to.
- That burning never reopens mint capacity.
- That redemption cannot be paused.

THE HOLDER-FACING VIEW

3 The economic loop

The contract system has many boundaries. The loop a holder actually participates in has five moves.

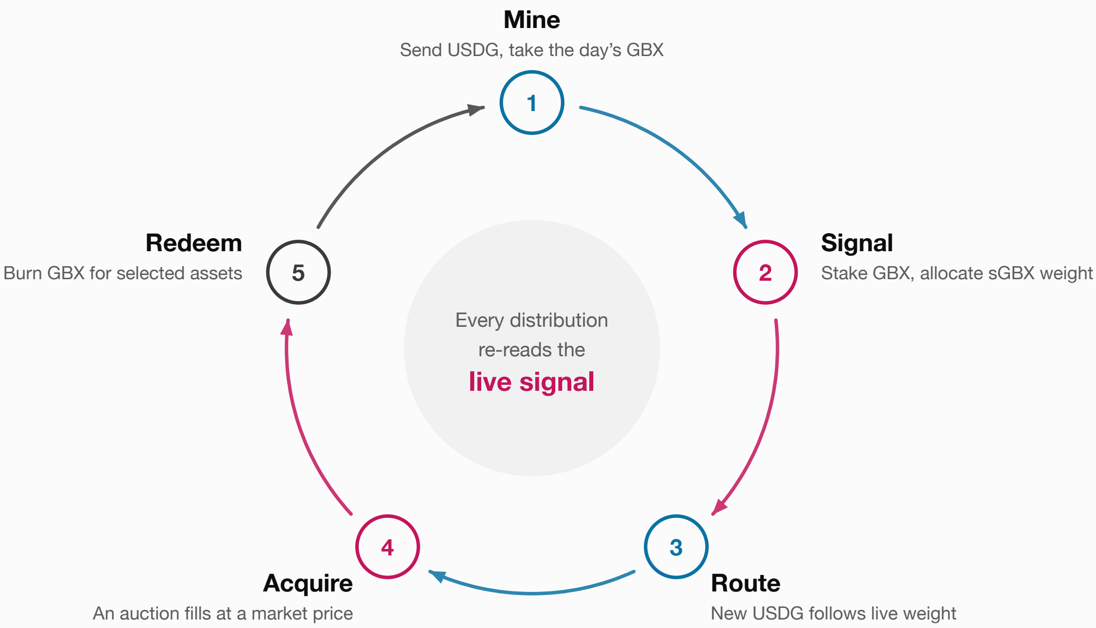


FIG. 03 The loop has no start state to wait for and no round to synchronise with. Mining, signaling, routing, acquiring and redeeming all proceed independently and continuously.

Revenue takes one path. A contribution reaches the Fundraiser, is forwarded to the single revenue entrance at **ResonanceRouter**, and arrives at **Resonance**, which distributes it among active Strategies in proportion to current sGBX weight.

The routing layer never forms an opinion. It does not price assets, rank Strategies, or hold a view about what deserves capital. It reads the live signal distribution and applies it to whatever arrived.

ONE ENTRANCE

Fundraiser contributions are the only USDG revenue entering this router. Liquidity-position fees belong entirely to the permissionless compounding incentive.

No proposal season. No signaling epoch. No withdrawal lock on unallocated sGBX. Every distribution re-reads the signal that exists at that moment.

SIX CONCEPTS CARRY THE ECONOMICS

4 Protocol components

Implementation detail stays behind narrow contract boundaries. A holder can reason about the protocol through six pieces.

PIECE	ROLE
USDG	The stable asset contributed by miners and routed toward acquisitions.
GBX	The transferable protocol token, capped at one billion cumulative mints, redeemable against Fund assets.
sGBX	Non-transferable staked GBX. It measures an account's live signal weight and nothing else.
Strategy	An eligible destination for USDG: normally an asset acquisition, or a GBX buyback.
Resonance	The allocation engine that distributes newly received USDG according to current signals.
Fund	The permissionless raw-token treasury that holds acquired assets and pays in-kind redemptions.

WHY SGBX CANNOT MOVE

Non-transferability keeps signal weight attached to someone who is exposed to the outcome. A transferable signal token would become a market in influence, rentable by anyone with no position in the Fund.

WHY THE FUND IS NOT A REGISTRY

The Fund accepts any ERC-20. Membership of the protocol is expressed by active Strategies, not by a curated list inside the treasury. Section 11 covers what that admits.

Those six pieces are carried by twelve direct, non-upgradeable contracts. The extra names exist to keep each responsibility narrow enough to review in isolation; none of them adds a decision.

The twelve contracts, grouped by what they are for

TOKENS	DISTRIBUTION	ROUTING	ACQUISITION AND REWARDS
`GBX`	`Fundraiser`	`ResonanceRouter`	`Strategy`
`SignalGBX`	`LiquidityPosition`	`Resonance`	`BribeRouter`
		`StrategyFactory`	`Bribe`
		`BribeFactory`	`Fund`

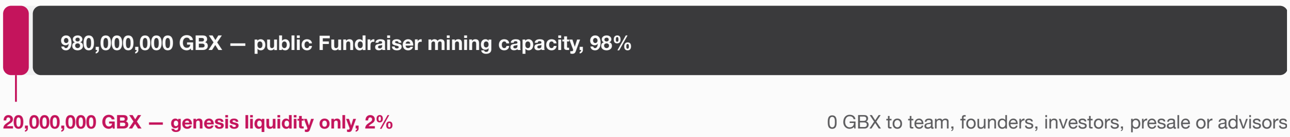
FAIR LAUNCH THROUGH THE FUNDRAISER

5 GBX mining and supply

Mining here means contributing USDG through the public Fundraiser and receiving that period's scheduled GBX. It is not proof-of-work.

LIFETIME MINT CEILING

1,000,000,000 GBX



Contributions within a day share that day's emission in proportion to their size. An empty day forfeits its scheduled emission entirely; nothing carries forward. The schedule's two constants are not independent: the daily decay is the 1,460-day half-life factor, and the opening emission is derived from it so the schedule pays out its own allocation and no more.

THE SCHEDULE FUNDS ITSELF EXACTLY

$$E_0 = A \times (1 - d), \quad \text{so} \quad \sum_{t\geq 0} E_0 \cdot d^t = E_0 / (1 - d) = A$$

A = 980,000,000 GBX mining allocation · d = 0.999525354337060160 · E_0 = 465,152.749681042811702004 GBX. Summing the entire infinite schedule returns the allocation exactly, so the cap is a property of the arithmetic rather than a separate check.

BURNING IS ONE-WAY

Burning GBX never restores mint capacity. Cumulative minting can never exceed one billion even if every previously minted token has since been destroyed.

FORFEITURE IS REAL

A day with no contributions emits nothing, and that emission is gone. The schedule advances on wall-clock time, not on activity.

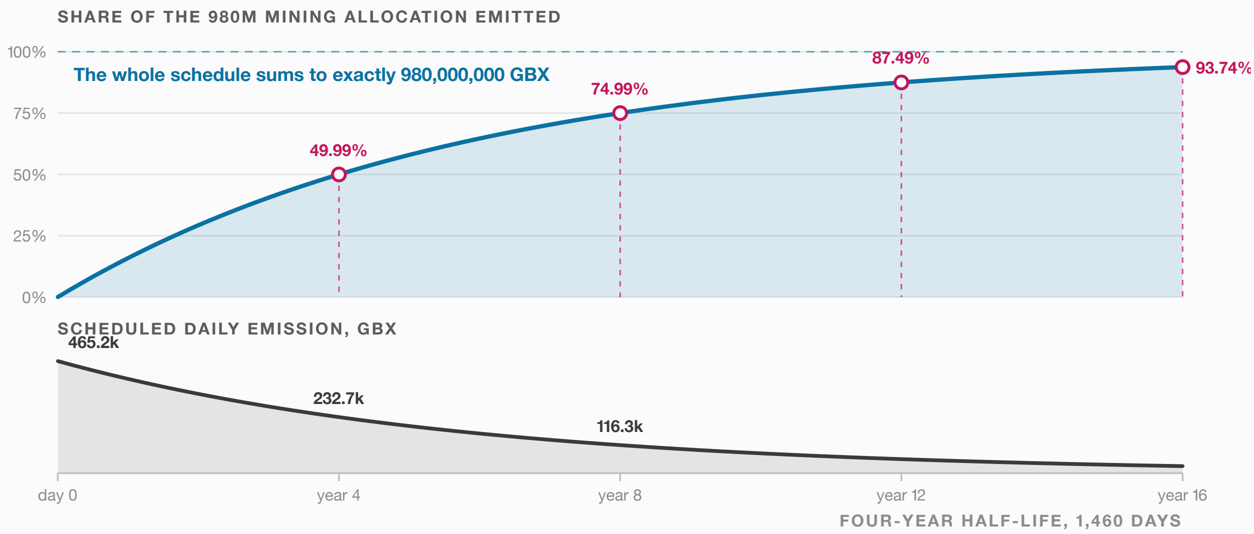


FIG. 04 Each four-year period closes half the remaining distance to the allocation: 50% by year four, 75% by year eight, 93.75% by year sixteen. The daily emission never reaches zero and the total never exceeds 980M.

WHY A LIQUID GBX PRICE PULLS CAPITAL IN

6 The mining market

The Fundraiser never sells GBX at a price. It hands a fixed daily quantity to whoever showed up, in proportion to what they brought.

PRO-RATA DAILY MINING

$$GBX_i = \mathrm{floor}(E_t \times c_i / C_t)$$

E_t = GBX emitted on day t · c_i = miner i's USDG contribution · C_t = all USDG contributed that day.

READ THE CHART THIS WAY

The rising line is what miners pay on average. The dashed line is what GBX is worth. The wedge between them is the margin that attracts the next miner — and closes as they arrive.

Because the numerator is fixed, the average cost a day's miners pay is simply C_t / E_t — it depends only on how much total USDG competed, never on who contributed it. If GBX trades at P_t, the day's emission is worth $P_t \times E_t$, and entry is attractive while contributions sit below that. Entry then closes the gap, which gives a rough break-even benchmark of $C_t \approx P_t \times E_t$. The adjustment runs both ways: a thin day is precisely the day worth entering.

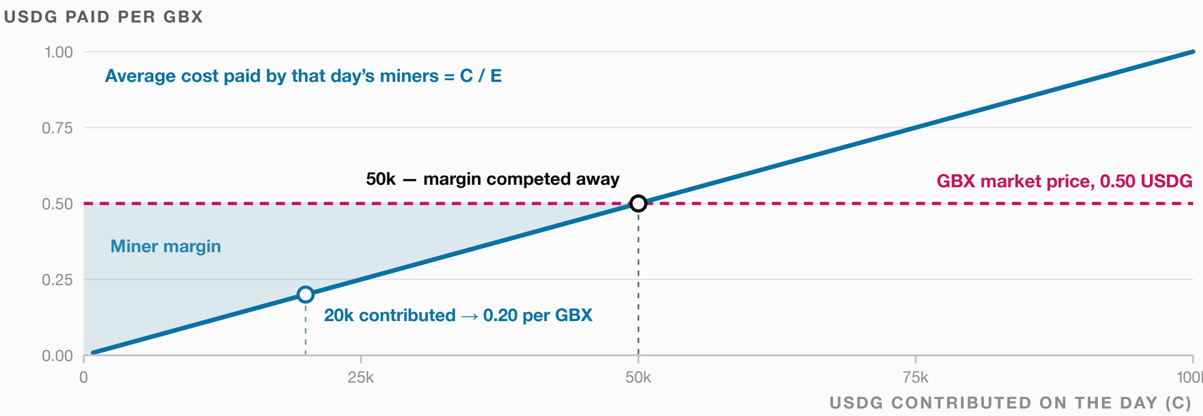


FIG. 05 An illustrative day: E = 100,000 GBX emitted, P = 0.50 USDG. At 20,000 USDG contributed the average cost is 0.20 per GBX; competition pushes toward 50,000 USDG, where the simple spot-price margin is gone. Entering below the dashed line is profitable only before gas, slippage and risk.

WHAT THE CONTRACTS GUARANTEE

- Every contributed USDG enters the signal-directed path, never a discretionary wallet.
- Miners compete for a fixed schedule, not a team-selected sale price.
- The Fund keeps the majority of every acquisition, whatever the share is set to.

WHAT NO CONTRACT CAN GUARANTEE

- That a quoted GBX price survives slippage, or that liquidity exists to realize it.
- That miners participate at all on a given day.
- That auctions clear, or that Fund assets hold value.

CHOOSE WHAT YOU WANT TO ACCUMULATE

7 Signaling with sGBX

Stake GBX one-for-one into non-transferable sGBX, then point that weight at the Strategies whose assets you actually want.

ADD ABSOLUTE AMOUNTS; LEAVE ANY REMAINDER IDLE

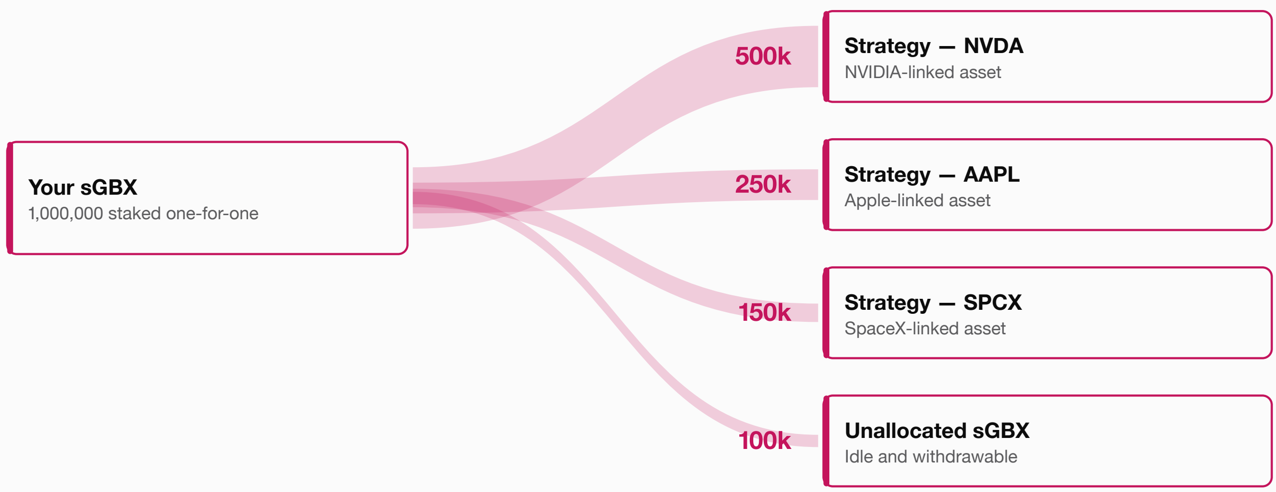


FIG. 06 Absolute amounts can be added to or removed from active Strategies independently — no epoch, cooldown or forced whole-account reset. Unallocated sGBX is immediately withdrawable. Changing an allocation redirects the next distribution; it never sells what the Fund already holds.

If the active Strategies target an NVIDIA-linked asset, an Apple-linked asset, a SpaceX-linked asset or anything else eligible, a holder signals the Strategy for the asset they want the protocol to accumulate. When that Strategy completes an acquisition, the signal-reward share is streamed in *that acquired asset* across eligible signal balances — so the reward arrives in the thing the signaler asked for.

Signaling is continuous and reversible. Each operation adds or removes an absolute per-Strategy amount; the amount is a delta, not a target. A holder may leave some sGBX idle and withdraw that unallocated balance immediately. Idle sGBX earns nothing and dilutes nothing.

A SIGNAL IS NOT A CLAIM

Signaling alone earns nothing. The Strategy must receive capital, and an auction must actually complete, before there is any asset to stream. Rewards depend on eligible weight at the moment of distribution.

AND NOT A VOTE

There is no threshold to cross and no outcome to lose. A 1% signal directs roughly 1% of the next distribution, whatever everyone else does.

THE NEXT DISTRIBUTION FOLLOWS LIVE WEIGHT

8 Capital allocation

Revenue is divided twice on its way to the Fund. Only the first division responds to holders.

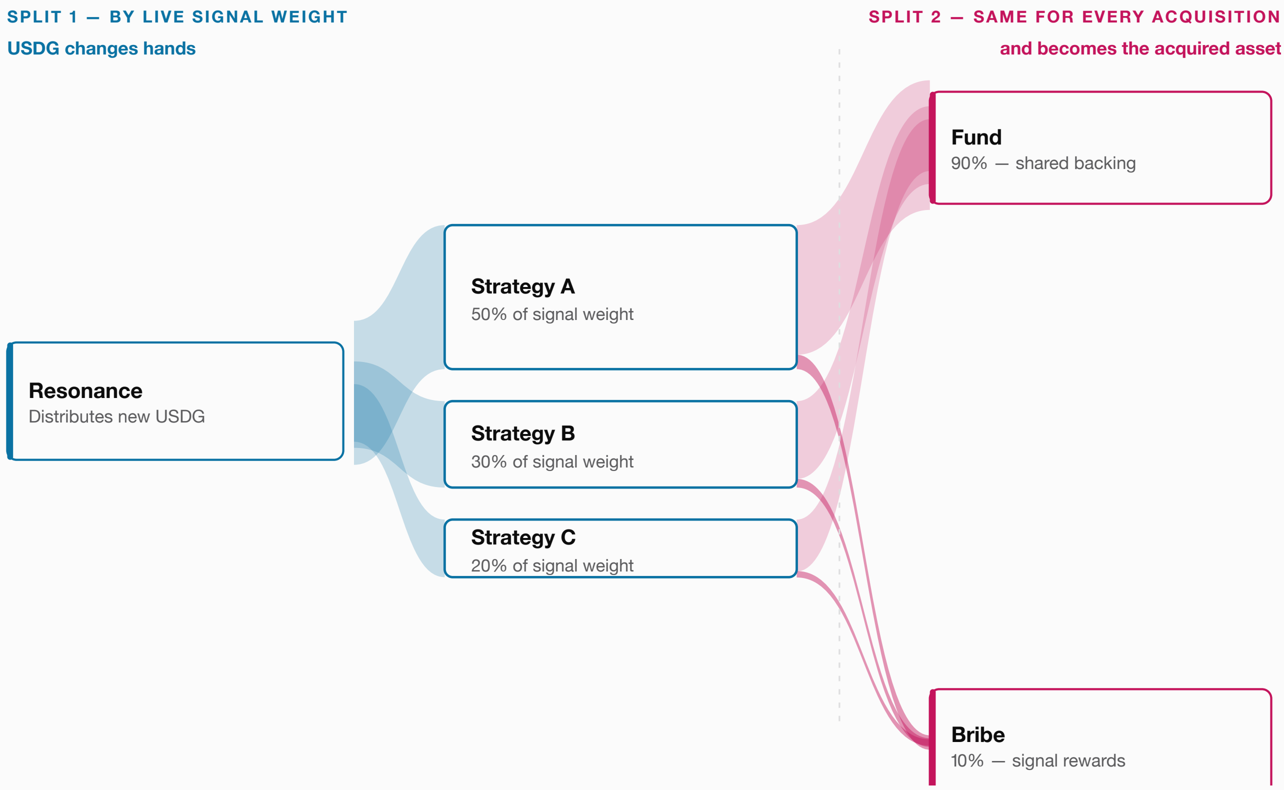


FIG. 07 Ribbon thickness is proportional to share. The first split is whatever holders currently want; the second is the same for every acquisition the protocol will ever make.

ALLOCATION TO STRATEGY I

```
allocationᵢ = floor(R × Sᵢ / S_total)
```

R = revenue being distributed · S_i = active signal weight on Strategy i · S_total = total active signal weight. Rounding and no-signal behaviour are implementation details that must preserve value and avoid discretionary routing.

ORDER MATTERS

Capital is split by signal *before* anything is acquired, and by the reward share *after*. No party sits between the two splits with a choice to make.

A Strategy holding 30% of active signal weight receives about 30% of the next distribution. Signals apply when new revenue arrives and only then, which is why a change of mind is cheap and a change of holdings is not.

ORACLELESS REVERSE DUTCH AUCTIONS

9 Acquisitions

A Strategy holds USDG and wants an asset. Rather than pricing that asset, it lets the clock lower its demands until someone accepts.

The direction confuses people, so state it plainly: the Strategy is the *seller of USDG*. It offers its accumulated USDG balance and asks a filler to hand over a quantity of the target asset in exchange. The auction opens by demanding a large quantity — a good deal for the protocol and a bad one for the filler — and that demand decays linearly to zero as the epoch elapses.

WHO PAYS WHOM

Filler → Strategy: the target asset. Strategy → filler: its USDG balance. Both legs settle in one transaction, so neither side is exposed to the other.

REQUIRED PAYMENT DURING AN EPOCH

```
payment(t) = init − floor(init × t / period)
```

There is no floor price. If nobody fills, the requirement reaches zero at the end of the epoch. The moment someone chooses to fill is itself the price discovery.

ONE EPOCH — TARGET ASSET THE FILLER MUST PAY

FOUR EPOCHS IN SEQUENCE

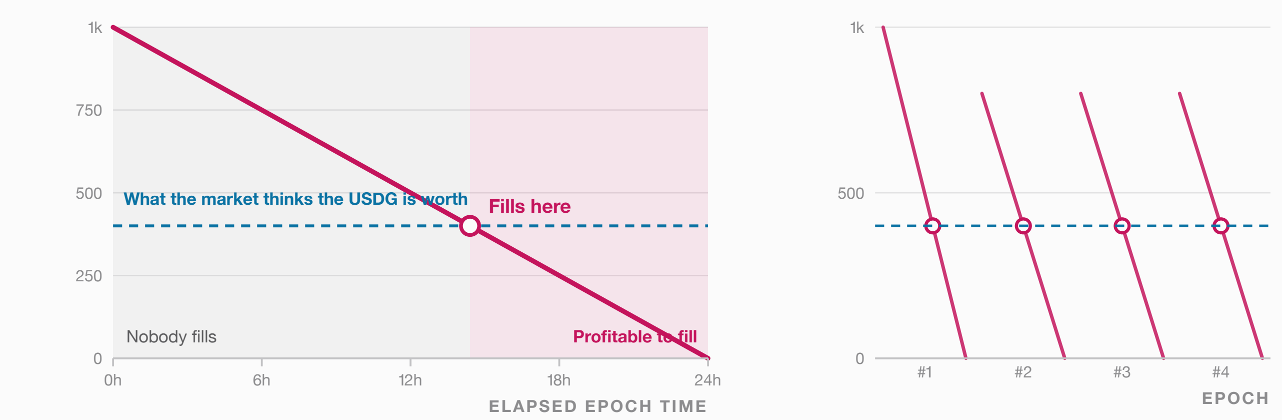


FIG. 08 Left: the requirement decays linearly to zero, and a rational filler waits until it falls below what the USDG is worth to them — the fill time is the price. Right: each fill re-opens the next epoch at the fill price multiplied by a bounded factor, so the sequence hunts the market price and re-tests it every epoch.

That ratchet is what removes the oracle. The protocol never reads a price; it proposes one, observes whether the market takes it, and adjusts. A fill that comes early says the opening ask was too generous, and the next epoch opens higher. A fill that comes late, or not at all, says the opposite.

This removes an oracle dependency, not market risk. Thin liquidity, unusual token behaviour, low participation or badly chosen parameters can still produce delayed or unfavourable execution.

BOUNDED BY CONSTRUCTION

Epoch length is bounded between one hour and one year, and the price multiplier between 1.1× and 3×. The bounds constrain how fast the ratchet can move, not where it settles.

MOST BUILDS SHARED BACKING; SOME REWARDS THE SIGNAL

10 Fund formation and signal rewards

Every completed normal acquisition does two things at once: it strengthens the common basket, and it pays the holders who directed it.

The acquired asset splits on a fixed rule. Ninety percent enters the Fund and becomes shared raw-token backing available to every GBX holder through proportional in-kind redemption. Ten percent funds the Strategy's Bribe, which streams that same asset across eligible signal balances.

Ten percent is the launch value, not a constant. The share is adjustable through timelocked governance and bounded: it may never exceed 50%, so a majority of every acquisition always reaches the Fund. Raising it moves value from shared backing toward the holders who directed that specific acquisition; lowering it does the reverse. If a Strategy has no eligible signalers at the moment of distribution, the reward share does not accumulate anywhere or wait for a claimant — it returns to the Fund.

The structure pairs a common benefit with a personal one. Most of what a signal directs becomes backing for everyone; the smaller share lets the holder who chose it hold the thing they chose.

STREAMED, NOT AIRDROPPED

Rewards accrue against signal weight held over the distribution period, using a reward-per-weight accumulator. Signaling immediately before a fill does not retroactively earn the whole reward.

BUYBACKS PAY NOTHING

A buyback Strategy has no acquired asset to split. It burns everything it buys, so there is no reward leg at all.

> **Signal the asset you want, and a completed acquisition pays you in that asset — while the larger share joins the basket every GBX holder can redeem against.**

Why not pay rewards in GBX?

Paying in the acquired asset keeps the incentive aligned with the choice. A holder who signals an NVIDIA-linked Strategy because they want that exposure receives that exposure. Paying in GBX would convert every signal into a claim on the protocol token regardless of what was actually bought, and would make signaling a yield strategy rather than a statement of preference.

BURN GBX, CHOOSE ASSETS, RECEIVE YOUR SHARE

11 The Fund and redemption

The Fund is permissionless raw-token custody, not a curated registry. The redeemer, not the protocol, decides which of its assets to claim.

PAYOUT FOR EACH SELECTED ASSET

$$\mathrm{payout}_j = \mathrm{floor}(\mathrm{balance}_j \times \mathrm{burned} \, / \, \mathrm{supplyBeforeBurn})$$

Supply and balances are snapshotted before the burn. The burn and every selected transfer are atomic: if one selected transfer fails, the entire redemption reverts, including the burn.

REDEMPTION CANNOT BE PAUSED

There is no pause switch anywhere in the core, which means there is none on the exit either.

BURN 1,000 GBX WHILE SUPPLY IS 100,000 GBX — A 1% CLAIM

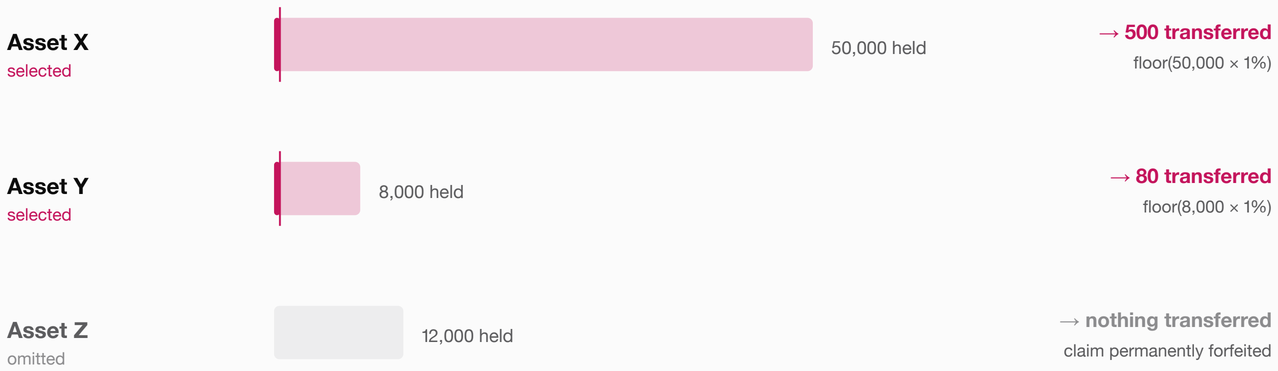


FIG. 09 Selecting an asset claims a pro-rata slice of it. Omitting one forfeits that claim permanently — the value stays behind for whoever still holds GBX.

Selectivity is the design's answer to a real hazard. Because the Fund accepts any ERC-20, it can end up holding tokens that are malicious, frozen, rebasing, fee-on-transfer, or simply broken. If every redemption had to touch every asset, one hostile token would be enough to disable the exit for everyone.

Letting the caller name the assets removes that dependency, at a price the redeemer pays knowingly: an omitted claim is gone, not deferred.

MEMBERSHIP LIVES ELSEWHERE

Official protocol membership is represented by active Strategies in Resonance, not by an asset list inside the Fund. Anything can arrive in the Fund; only a Strategy can direct capital at it.

ONE MARKET LOOP, ONE SUPPLY-REDUCTION LOOP

12 Liquidity fees and buybacks

The canonical market position auto-compounds; the buyback Strategy terminates in an atomic burn.

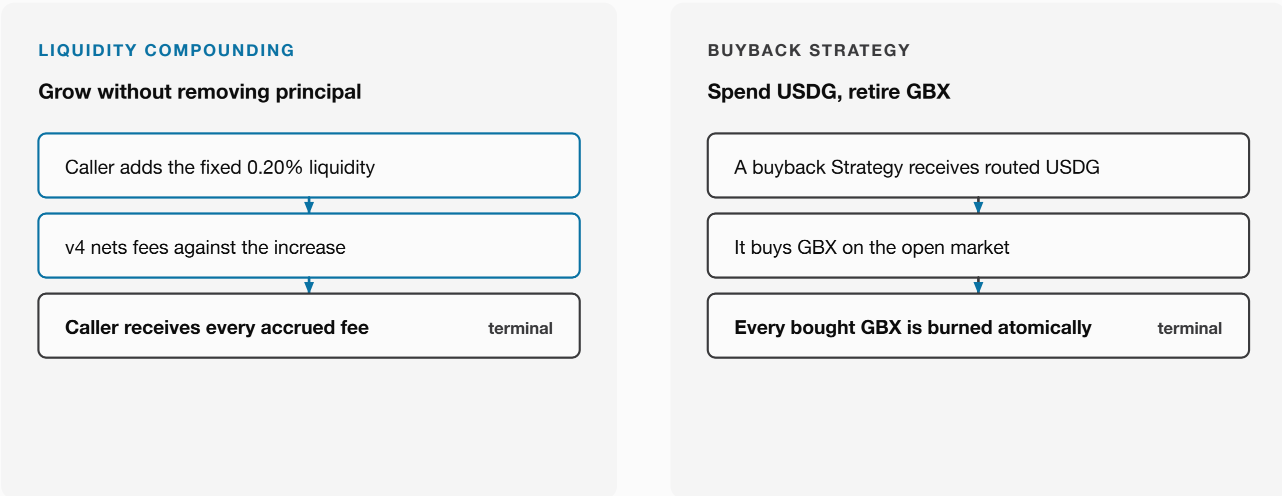


FIG. 10 The position grows by a fixed rule and pays its caller accrued fees; buybacks destroy every GBX payment.

The canonical market position is a precommitted, hookless GBX/USDG Uniswap v4 position funded with the 20 million GBX genesis allocation. Compounding is permissionless and removes zero principal: a caller grows liquidity by the fixed 0.20% and receives every accrued fee. Position fees are not protocol revenue.

A buyback Strategy is an ordinary Strategy whose target asset is GBX itself. It spends routed USDG, buys GBX on the open market, and burns everything it receives, atomically. It pays no signal reward and must never leave purchased GBX sitting as inventory.

GENESIS LIQUIDITY IS NOT A TREASURY

The 20M genesis tranche exists to make GBX tradeable. Its position NFT cannot be withdrawn to an arbitrary receiver, and the fees it generates never return to an operator.

WHY A LIQUID MARKET MATTERS

Section 6 depends on GBX having a realizable price. Genesis liquidity is what makes that price exist at all, which is why it is committed rather than allocated.

SIGNALS STEER FLOW, NOT HOLDINGS

13 Why the basket is path-dependent

Because every distribution follows the signal that exists at that moment, the Fund ends up holding the sum of every past decision rather than the current one.

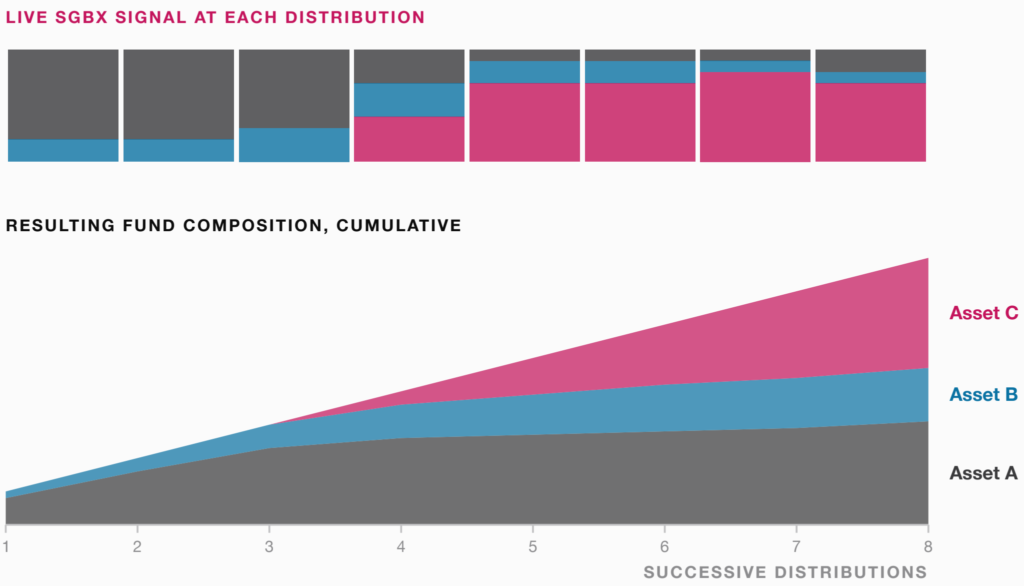


FIG. 11 The upper band is what holders want at each distribution. The lower area is what the Fund actually holds. They are related by accumulation, and they are never the same picture.

This is the most important behavioural consequence in the design, and the one most likely to be misread. A holder who moves signal to a new Strategy has changed the destination of future capital and nothing else. The assets bought under the old signal remain in the Fund until someone redeems them out.

Composition therefore reflects the full history of contributions, signals and completed acquisitions. Two protocols with identical signal distributions today can hold entirely different baskets, because they arrived by different routes. There is no rebalancing mechanism that reconciles the two, and adding one would require exactly the discretionary selling power the design removes.

WHAT SPEEDS IT UP

Only new revenue. A basket shifts faster when contributions are large relative to existing holdings, which means early distributions shape composition far more than later ones.

WHAT CAN UNWIND IT

Redemption. A holder who burns GBX and selects only some assets removes those specific holdings from the Fund, which is the one mechanism that reduces an existing position.

A WORKED EXAMPLE

14 One contribution, end to end

Everything above, applied once, to 1,000 USDG. Figures are illustrative and assume each step completes.

1. **Contribute 1,000 USDG**
   On a day scheduled to emit 100,000 GBX, where 50,000 USDG is contributed in total.
   You receive 2,000 GBX — a 2% share of the day.

2. **Stake it into 2,000 sGBX**
   You allocate all of that weight behind the NVDA Strategy.
   Your weight now steers your slice of every future distribution.

3. **Your 1,000 USDG is routed**
   It reaches ResonanceRouter, then Resonance, which reads live signal weights.
   If NVDA holds 30% of signal weight, that Strategy receives about 300 USDG.

4. **The Strategy runs an auction**
   Its accumulated USDG is offered, and the required NVDA payment decays until a filler acts.
   The Strategy receives NVDA-linked tokens at a market-set rate.

5. **The acquisition splits**
   At the current signal-reward share, which starts at 10% and is capped at 50%.
   90% joins the Fund; 10% streams to eligible NVDA signalers, including you.

6. **Later, you redeem**
   You burn GBX and name the Fund assets you actually want.
   You receive a pro-rata share of exactly those assets, in kind.

FIG. 12 Each step is a separate transaction and none of them is guaranteed to follow the last. The Strategy may not fill; the signal may be reallocated; the holder may never redeem.

Two details are worth pausing on. First, the 2,000 GBX in step 1 depends on what everyone else contributed that day, not on any price the protocol set — a quieter day would have produced more GBX for the same 1,000 USDG. Second, the routing in step 3 sends capital wherever signal weight points at that moment, which is not necessarily where this particular contributor pointed it.

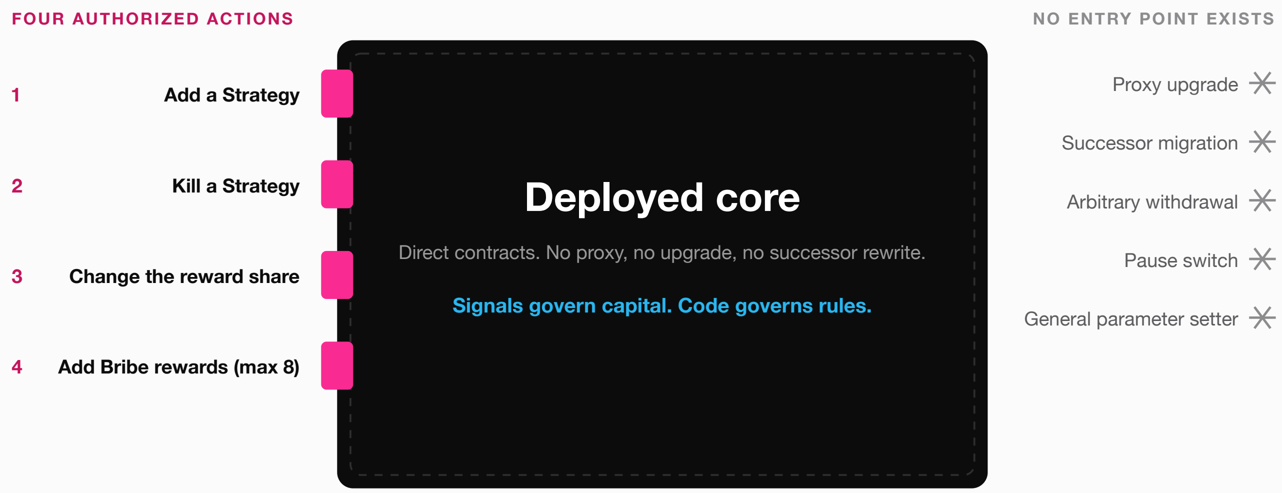
YOU ARE NOT BUYING YOUR OWN SIGNAL

Your contribution joins a common pool and follows the aggregate distribution. Signaling raises the share of *all* revenue that reaches your chosen Strategy, including everyone else's.

SIGNALS GOVERN DIRECTION; THE CORE STAYS FIXED

15 Governance-minimized final design

Economic direction and system maintenance are deliberately separated, and only one of them is a human decision.



Immutability removes the power to fix mistakes as surely as it removes the power to cause them. That is the trade, stated plainly.

FIG. 13 The four authorized actions are the entire management surface. Everything on the right has no entry point, not a restricted one.

sGBX holders direct capital continuously by signaling active Strategies. Management maintains the edges of the system — which Strategies exist, how each acquisition divides between the Fund and its signalers, and which additional Bribe rewards are registered — and can do nothing else. Each Bribe is permanently capped at eight reward tokens.

Open work: whether multi-token Bribe rewards should exist at all remains unresolved. The eight-token cap bounds the current path without deciding that product question.

MINIMIZED, NOT ABSENT

A compromised manager key can still add or kill Strategies, push the signal-reward share to its 50% cap, and register rewards up to the eight-token limit. It cannot upgrade the core, withdraw Fund assets, pause redemption, or touch what the Fund already holds.

PROPERTIES THE IMPLEMENTATION MUST MAKE TESTABLE

16 Core invariants

An invariant is a statement that should hold after every valid transaction, regardless of who calls it or in what order.

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01 Cumulative GBX minting never exceeds one billion.

02 Burning GBX never restores mint capacity.

03 Fundraiser emissions follow the fixed sequential daily schedule, and empty days forfeit without carry.

04 All contribution revenue enters the signal-directed routing path.

05 Absolute per-Strategy sGBX signals change by incremental deltas; unallocated sGBX is withdrawable.

06 Redemption uses pre-burn supply and balance snapshots.

07 A redemption burn and all selected transfers are atomic.

08 A normal acquisition divides the acquired asset at the current signal-reward share, which starts at 10% and can never exceed 50%, and returns that share to the Fund when no eligible signalers exist.

09 A buyback burns all received GBX atomically and pays no signal reward.

10 Each Bribe has at most eight append-only reward tokens.

11 Liquidity compounding adds 0.20% and removes no principal.

12 The deployed core has no upgrade path and no arbitrary asset-withdrawal path.

WHY THESE AND NOT OTHERS

Each one names a way value could leave the system unexpectedly, or a discretionary power that could be reintroduced by accident. They are chosen to be falsifiable by a test, not to be reassuring.

STATED, NOT PROVEN

This is a specification of intent. None of these properties is currently established by an independent audit of final bytecode.

ONCHAIN DOES NOT MEAN RISK-FREE

17 Risks and trust assumptions

Removing discretionary powers narrows what an operator can do. It does not narrow what a market, a token, or a bug can do.

WHERE	WHAT CAN FAIL	WHAT THE DESIGN DOES ABOUT IT
Smart contracts	A flaw breaks routing, auctions, rewards, burns or redemption.	Narrow contract boundaries and stated invariants. Immutability makes any surviving flaw permanent.
Manager key	A compromised key adds or kills Strategies, changes the reward share, or registers Bribe rewards.	Four bounded actions only. No upgrade, withdrawal, pause, or way to reach the Fund's existing holdings.
Signal concentration	A large sGBX holder directs a disproportionate share of future capital.	Nothing. Weight is proportional by design; concentration of GBX becomes concentration of direction.
Auction execution	Thin participation or bad parameters produce a poor fill, or no fill at all.	Bounded epoch and multiplier, and a ratchet that re-tests the market. Neither guarantees a counterparty.
Asset quality	The Fund receives malicious, frozen, rebasing or fee-on-transfer tokens.	Selective redemption means one broken token cannot disable the exit for everyone.
Selective redemption	A caller selects a broken token and reverts, or omits one and loses the claim.	The choice is explicit and per-call. The forfeited claim stays with the remaining GBX supply.
External systems	USDG, the chain, bridges, sequencers or custody fail.	Nothing. These are dependencies outside the core contracts and outside its guarantees.
Economics	Demand, participation, fills and asset values all disappoint.	Nothing is promised about any of them. Section 6 states the incentive and its conditions.
Legal and regulatory	Tokenized assets carry issuer, custody, transfer and jurisdictional restrictions.	Nothing. Holding a tokenized claim is not equivalent to owning the underlying asset.

A smaller management surface narrows what an operator can do to you. It cannot supply good assets, good prices, good signals, or bug-free code.

A SPECIFICATION, NOT A LAUNCH CLAIM

18 Implementation status

This paper describes a target final design. The repository behind it is a pre-audit engineering starting point.

The current contracts, generated interfaces, indexing layer, deployment scripts and technical documentation still describe a broader administrative surface than this paper specifies. Those differences must be removed and the whole repository reconciled before any deployment is considered.

Minimum requirements before production

01 Final contract interfaces matching the four-action Resonance management surface.

02 An owner decision on whether multi-token Bribe rewards should exist.

03 Complete unit, integration, invariant and economic-model tests.

04 Reviewed asset and chain configuration.

05 Reproducible deployment rehearsals.

06 Independent security review of the final immutable bytecode.

07 Explicit resolution of every open trust, licensing and operational question.

QUESTION STILL TO DECIDE

Bribe reward registration is bounded at eight tokens, but the stronger question remains: should additional reward tokens exist at all?

WHAT A GREEN BUILD MEANS

A passing local build is engineering evidence only. It is not an audit, an authorization, or a launch claim.

A PUBLIC MECHANISM FOR FORMING A BASKET

19 Conclusion

GumBall6900 turns fund formation into a continuous onchain process instead of publishing a finished list.

GBX is a transferable claim, redeemable against assets the Fund already holds. sGBX is a live statement about where arriving capital should go. Strategies convert that statement into acquisitions, and the accumulated history of those acquisitions is the basket.

THE NARROW CLAIM

Not that holders will choose well. Only that the choosing can be public, continuous, and inspectable.

	REFERS TO	CHANGES WHEN
GBX	What the Fund already owns	Someone mines, burns, or redeems
sGBX	What the Fund may acquire next	A holder reallocates weight — instantly
The fixed core	The rules both must follow	Never, after deployment

Public mining. Continuous signals. Market-executed acquisitions. In-kind redemption. A fixed core.

The protocol does not promise that holders will select good assets, that auctions will clear, or that GBX will retain value. Its proposition is narrower than any of those: that public mining, continuous signaling, market-executed acquisition, in-kind redemption and a governance-minimized core can make shared capital allocation more open and more inspectable than the alternative.

REFERENCE

Glossary

Short definitions for the terms this paper relies on.

Bribe	The Strategy-specific reward contract that streams an acquired asset across eligible signal balances.
Eligible signaler	An account whose sGBX weight was allocated to a Strategy over the period a reward accrued.
Fund	The permissionless raw-token treasury that holds acquired assets and supports selective in-kind redemption.
Fundraiser	The public USDG contribution mechanism through which GBX is mined on a fixed daily schedule.
GBX	The transferable protocol token, capped at one billion cumulative mints.
Genesis liquidity	The 20 million GBX committed to the canonical GBX/USDG position. Not a treasury and not withdrawable to an arbitrary receiver.
In-kind redemption	Receiving the underlying tokens themselves rather than a cash or synthetic equivalent.
Bribe reward cap	At most eight append-only reward tokens may be registered for one Strategy.
Mining	Contributing USDG through the public Fundraiser and receiving that period's scheduled GBX. Not proof-of-work.
Path-dependent	A composition determined by the order and timing of past events, not by current preferences alone.
Resonance	The allocation engine that distributes incoming USDG according to live sGBX signals.
Reverse Dutch auction	An auction whose asking amount falls over time. Here the Strategy offers USDG and the target-asset payment it requires decays to zero.
sGBX (SignalGBX)	Non-transferable staked GBX used as live signal weight.
Signal weight	The share of active sGBX allocated to a Strategy, which determines its share of the next distribution.
Strategy	An active, eligible acquisition or buyback path registered in Resonance.
USDG	The stable asset contributed by miners and routed through the protocol.

DOCUMENT BASIS

If this paper and the deployed bytecode ever disagree, the bytecode wins.

This paper summarizes the intended target design recorded in the GumBall6900 repository as of 8 August 2026. The technical source of truth remains the final reviewed contracts and their matching specifications. Prose is maintained in `docs/WHITEPAPER.md`; this typeset edition is generated from `docs/whitepaper`.

Charts are computed at build time from the same emission, auction and redemption rules the repository's reference model tests, so no figure in this document contains a hand-transcribed number.

COLOUR KEY

USDG capital · Signal, and what it buys · GBX supply and burns

EDITION

v0.2 · August 2026

STATUS

Pre-audit. Not deployed. Not authorized for user funds.

DISCLOSURE

Educational protocol overview only. Not investment, legal, or tax advice.